NBA Project

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August 2025

1 Introduction

Scholars and commentators frequently note a persistent preference within U.S. cultural industries for domestically born talent over foreign counterparts. This tendency appears across multiple domains. In film, American actors dominate marquee productions and typically command greater media visibility. In popular music, American artists often receive preferential coverage and award recognition. Such patterns are commonly interpreted as reflecting an implicit hierarchy of cultural and professional value that privileges "local" status. The dynamic is especially salient in professional sports—and most notably in the National Basketball Association (NBA), widely regarded as the premier global basketball league. While international players increasingly achieve on-court success and commercial prominence, perceptions of American players often remain comparatively elevated, sustained by tacit assumptions of superior fit, professionalism, or cultural alignment. This paper examines whether and how such preferences manifest in measurable outcomes in the NBA, and to what extent they persist after controlling for observable performance.

2 The Data

The empirical analysis draws upon a publicly available dataset retrieved from the Kaggle platform, encompassing individual performance statistics for all NBA players during the 1996–2022 seasons. The dataset includes a broad set of advanced and traditional basketball metrics, from which the following were selected for this study:

- **Possessions** (`possession`) — The count of discrete offensive sequences initiated by the player's team, each constrained by the 24-second shot clock.
- **Games Played** (`gp`) — Total number of games in which the player appeared during the season.
- **Points per Game** (`pts`) — Mean scoring output per game.

- **Rebounds per Game** (`reb`) — Mean number of rebounds secured per game, defined as gaining possession following a missed field goal attempt.
- **Assists per Game** (`ast`) — Mean number of assists per game, awarded when a pass directly leads to a made basket.
- **Net Rating** (`net_rating`) — The net point differential per 100 possessions while the player is on the court, calculated as team points scored minus points allowed.
- **Usage Percentage** (`usg_pct`) — The proportion of team offensive possessions that involve the player, through shot attempts, free throw attempts, or turnovers, while the player is on the floor.
- **True Shooting Percentage** (`ts_pct`) — A composite efficiency metric integrating field goals, three-pointers, and free throws into a single shooting performance indicator.

This set of variables allows for an assessment of both quantitative output (e.g., points, rebounds, assists) and contextual performance metrics (e.g., net rating, usage, efficiency), enabling a balanced evaluation of player contributions.

3 Research Question and Hypotheses

To investigate this issue empirically, we compiled a dataset covering 26 NBA seasons (1996–2022), excluding players who participated in fewer than 20 games in a given season. The filtered dataset comprises 10,720 players, of whom 8,872 ($\approx 82.8\%$) were American and 1,848 ($\approx 17.2\%$) were international.

Statistical hypotheses:

- H_0 (Null Hypothesis): No statistically significant difference exists in performance between American and international players; thus, no discrimination is present.
- H_1 (Alternative Hypothesis): American players exhibit significantly higher performance measures, consistent with discriminatory preference against international players.

4 Methods

4.1 Discrimination Index: Construction and Inference

Definition. To operationalize potential discrimination, we construct a player-level index that compares a player's measured *contribution* to the *credit* he receives. Let i index players and t seasons. We first standardize each raw metric within season to ensure comparability across years:

$$\tilde{x}_{it} = \frac{x_{it} - \mu_{x,t}}{\sigma_{x,t}},$$

where $\mu_{x,t}$ and $\sigma_{x,t}$ are the season-t mean and standard deviation of metric x.

The *contribution* component aggregates production and impact:

$$C_{it} \;=\; w_{\text{pts}}\,\tilde{\text{pts}}_{it} \;+\; w_{\text{reb}}\,\tilde{\text{reb}}_{it} \;+\; w_{\text{ast}}\,\tilde{\text{ast}}_{it} \;+\; w_{\text{net}}\,\tilde{\text{net_rating}}_{it},$$

with $w_{\text{pts}} > w_{\text{reb}} > w_{\text{ast}} > w_{\text{net}}$ and $w_{\text{pts}} + w_{\text{reb}} + w_{\text{ast}} + w_{\text{net}} = 1$. (Weights are set *a priori* to reflect conventional importance in basketball, assigning the highest weight to scoring, followed by rebounding, assists, and on-court net rating.)

The *credit* component captures opportunity and role:

$$K_{it} \;=\; \alpha\,\tilde{\text{gp}}_{it} \;+\; \beta\,\tilde{\text{usg_pct}}_{it},$$

where gp is games played and usg_pct is usage percentage while on court. The coefficients $\alpha, \beta > 0$ scale opportunity (availability) and tactical involvement.

Our *Discrimination Index* is then

$$D_{it} \;=\; \frac{C_{it}}{K_{it}}.$$

Values $D_{it} > 1$ indicate contribution that exceeds the player's observed credit, while $D_{it} < 1$ indicates the reverse.

4.2 Normality Assessment: Shapiro–Wilk and Q–Q Plots

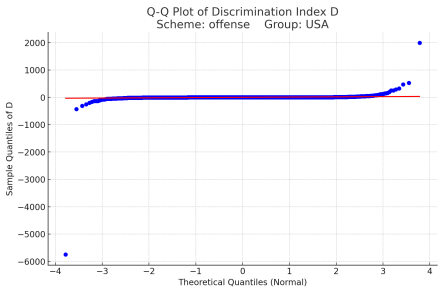
To determine whether parametric assumptions were appropriate, we assessed the distribution of the Discrimination Index D separately for American and international players using the Shapiro–Wilk test. The null hypothesis is that the sample is drawn from a normal distribution; small p-values (e.g., $p < 0.05$) indicate evidence against normality. We complemented the test with Q–Q plots, which compare sample quantiles to the theoretical quantiles of a normal distribution.

Q–Q Plots. The following figures show Q–Q plots of D for each weighting scheme, split by USA vs. International groups.

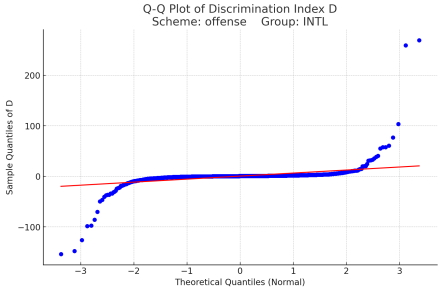
4.3 Group Comparison: Mann–Whitney U (Wilcoxon Rank-Sum) Test

Because D was non-normal and the two groups are independent (American vs. international), we used the one-sided Mann–Whitney U test to compare the distributions of D.

- **Idea:** Pool the two groups, rank all observations, and compare the sums of ranks. If one group tends to have larger values, its rank sum will be higher.

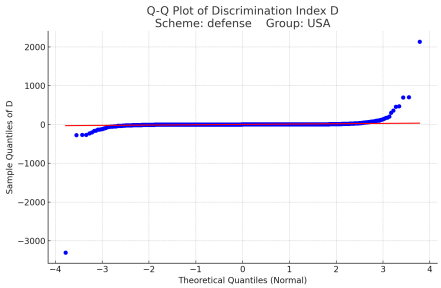


(a) Offense weights — USA

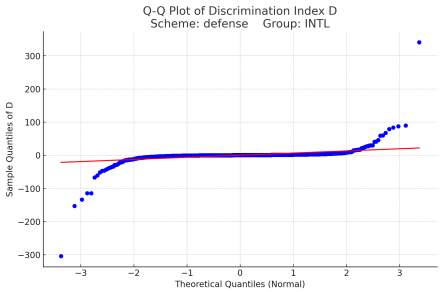


(b) Offense weights — INTL

Figure 1: Q–Q plots of the Discrimination Index D under offense-oriented weights.

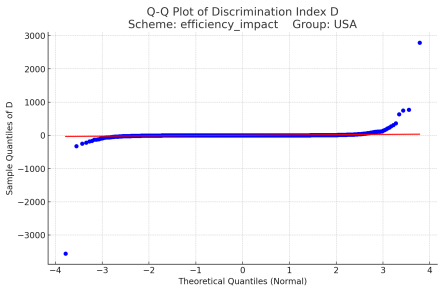


(a) Defense weights — USA

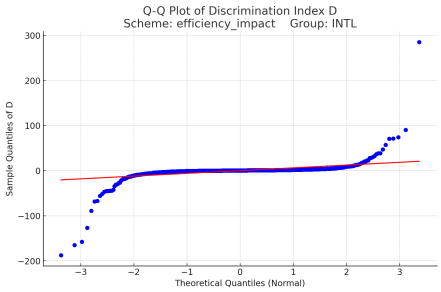


(b) Defense weights — INTL

Figure 2: Q–Q plots of D under defense-oriented weights.



(a) Efficiency/impact weights — USA



(b) Efficiency/impact weights — INTL

Figure 3: Q–Q plots of D under efficiency/impact weights.

- **Test statistic:** Let group 1 have size n_1 with rank sum R_1, and group 2 size n_2. Then

$$U_1 \;=\; n_1 n_2 \;+\; \frac{n_1(n_1+1)}{2} \;-\; R_1, \qquad U_2 \;=\; n_1 n_2 - U_1.$$

  Under H_0 (no shift between distributions), $\mathbb{E}[U_1] = \frac{n_1 n_2}{2}$ and $\mathrm{Var}(U_1) = \frac{n_1 n_2 (n_1+n_2+1)}{12}$ (with standard tie correction in practice). For large samples we use a normal approximation with a Z-score and compute the p-value accordingly.

- **Assumptions:** independent samples; ordinal or continuous outcome. If group shapes are similar, the test can be interpreted as a difference in medians.

- **Effect size:** We report $r = Z/\sqrt{N}$ (where $N = n_1 + n_2$) to quantify magnitude in addition to p-values.

5 Results

We test group differences in the Discrimination Index using the one-sided Mann–Whitney U test with alternative $\mathrm{median}(D_{\mathrm{INTL}}) > \mathrm{median}(D_{\mathrm{USA}})$, where higher D indicates greater on-court contribution per unit of "credit" (games played and usage).

To probe robustness, we pre-specify three weight schemes that reflect different emphases:

- **Offense emphasis:** $w_{\mathrm{ast}} = 0.25$, $w_{\mathrm{pts}} = 0.25$, $w_{\mathrm{ts}} = 0.25$, $w_{\mathrm{reb}} = 0.15$, $w_{\mathrm{net}} = 0.10$.
  Result: $p = 8.88 \times 10^{-3}$. We reject H_0 at the 5% level, indicating higher D for international players under an offense-oriented specification.

- **Defense emphasis:** $w_{\mathrm{reb}} = 0.30$, $w_{\mathrm{net}} = 0.25$, and $w_{\mathrm{pts}} = w_{\mathrm{ast}} = w_{\mathrm{ts}} = 0.15$.
  Result: $p = 9.53 \times 10^{-6}$. Evidence strengthens substantially when emphasizing defensive orientation.

- **Efficiency/impact emphasis:** $w_{\mathrm{ts}} = 0.30$, $w_{\mathrm{net}} = 0.25$, $w_{\mathrm{pts}} = 0.20$, $w_{\mathrm{ast}} = 0.10$, $w_{\mathrm{reb}} = 0.15$.
  Result: $p = 2.69 \times 10^{-6}$, the smallest among the three, implying the strongest evidence when weighting efficiency and overall on-court impact more heavily.

Across all specifications, the null hypothesis of equal distributions is rejected, with direction consistent with higher D among international players. Interpreted through the construction of D, this pattern is consistent with international players providing greater measured contribution per unit of observed credit (games played and usage).